Finals Lab Task 3. Using Dax Time Intelligence Function Overview:

In this lab, you'll create measures with DAX expressions that involve time intelligence.
In this lab, learn how to:

- Use various time intelligence functions to manipulate filter context that specific concerns dates.

This lab should take approximately 15 minutes.

Get started

To complete this exercise, first open a web browser and enter the following URL to download the zip file:
Source_here: 04-time-intelligence
https://drive.google.com/drive/folders/1TjDULUivAp6IpXe8DEWQJNp8rxcd9ilv?usp=sharing

1. **Open the 06-Starter-Sales Analysis.pbix file.**
2. **Create a YTD measure**
   - In this task, you'll create a sales year-to-date (YTD) measure by using time intelligence functions.
   - In Power BI Desktop, in **Report view**, on **Page 2**, notice the matrix visual that displays various measures with years and months grouped on the rows.
   - Add a measure to the Sales table, based on the following expression, and formatted to zero decimal places**: (You may check the .txt file for the copy of code)**

Code

```
Sales YTD =
TOTALYTD(
    SUM(Sales[Sales]),
    'Date'[Date],
    "6-30"
)
```

The `TOTALYTD` function evaluates an expression—in this case the sum of the `Sales` column—over a given date column. The date column must belong to a date table marked as a date table.

The function can also take a third optional argument representing the last date of a year. The absence of this date means that December 31 is the last date of the year. For Adventure Works, June is in the last month of their year, and so "6-30" is used.

3. Add the **Sales field and the Sales YTD** measure to the matrix visual.
4. Notice the accumulation of sales values within the year.

Sum of Sales	Sales YTD
$16,429,043	**$16,429,043**
$489,328	$489,328
$1,540,072	$2,029,400
$1,166,332	$3,195,733
$844,833	$4,040,566
$2,325,755	$6,366,320
$1,703,435	$8,069,756
$713,230	$8,782,985
$1,900,794	$10,683,780
$1,455,280	$12,139,060
$883,011	$13,022,071
$2,269,720	$15,291,791
$1,137,252	$16,429,043
$27,979,780	**$27,979,780**

The `TOTALYTD` function performs filter manipulation, specifically time filter manipulation. For example, to compute YTD sales for September 2017 (the third month of the fiscal year), all filters on the `Date` table are removed and replaced with a new filter of dates commencing at the beginning of the year (July 1, 2017) and extending through to the last date of the in-context date period (September 30, 2017).

Many time intelligence functions are available in DAX to support common time filter manipulations.

5. Create a **YoY growth measure**

In this task, you'll create a sales YoY growth measure by using a variable.

Variables help you simplify the formula and are more efficient if using the logic multiple times within a formula. Variables are declared with a unique name, and the measure expression must then be output after the `RETURN` keyword. Unlike some other coding language variables, DAX variables can only be used within the single formula._

1. Add another measure to the `Sales` table, based on the following expression:

Code

```
Sales YoY Growth =
VAR SalesPriorYear =
    CALCULATE(
        SUM(Sales[Sales]),
        PARALLELPERIOD(
            'Date'[Date],
            -12,
            MONTH
        )
    )
RETURN
    SalesPriorYear
```

Copy the code from the 06 – snippets.txt file

> *The `SalesPriorYear` variable is assigned an expression that calculates the sum of the `Sales` column in a modified context. That context uses the `PARALLELPERIOD` function to shift 12 months back from each date in filter context.*

6. Add the Sales YoY Growth measure to the matrix visual.

7. Notice that the new measure returns BLANK for the first 12 months (because there were no sales recorded before fiscal year 2017).

8. Notice that the Sales YoY Growth measure value for *2018 Jul* is the sales value for *2017 Jul*.

Sum of Sales	Sales YTD	Sales YoY Growth
$16,429,043	**$16,429,043**	
$489,328	$489,328	
[illegible]	$2,029,400	
[illegible]	$3,195,733	
$844,833	$4,040,566	
$2,325,755	$6,366,320	
$1,703,435	[illegible]	
$713,230	[illegible]	
$1,900,794	[illegible]	
$1,455,280	[illegible]	
$883,011	[illegible]	
$2,269,720	$15,291,791	
$1,137,252	$16,429,043	
$27,979,780	**$27,979,780**	**$16,429,042.6**
$2,411,559	$2,411,559	$489,328.4
$3,615,914	$6,027,473	$1,540,072.02

> *Now that the "difficult part" of the formula has been tested, you can overwrite the measure with the final formula that computes the growth result.*

9. To complete the measure, overwrite the Sales YoY Growth measure with this formula, formatting it as a percentage with two decimal places:

Code

```
Sales YoY Growth =
VAR SalesPriorYear =
    CALCULATE(
        SUM(Sales[Sales]),
        PARALLELPERIOD(
            'Date'[Date],
            -12,
            MONTH
        )
    )
RETURN
    DIVIDE(
        (SUM(Sales[Sales]) - SalesPriorYear),
        SalesPriorYear
    )
```

Copy the code from the 06 – snippets.txt file

10. In the formula, in the RETURN clause, notice that the variable is referenced twice.

11. Verify that the YoY growth for *2018 Jul* is 392.83 percent.

Sum of Sales	Sales YTD	Sales YoY Growth
$16,429,043	**$16,429,043**	
$489,328	$489,328	
$1,540,072	$2,029,400	
$1,166,332	$3,195,733	
$844,833	$4,040,566	
$2,325,755	$6,366,320	
$1,703,435	$8,069,756	
$713,230	$8,782,985	
$1,900,794	$10,683,780	
$1,455,280	$12,139,060	
$883,011	$13,022,071	
$2,269,720	$15,291,791	
$1,137,252	$16,429,043	
$27,979,780	**$27,979,780**	**70.31%**
$2,411,559	$2,411,559	$\boxed{392.83\%}$
$3,615,914	$6,027,473	134.79%

> ! *The YoY growth measure identifies almost 400 percent (or 4x) increase of sales during the same period of the previous year.*

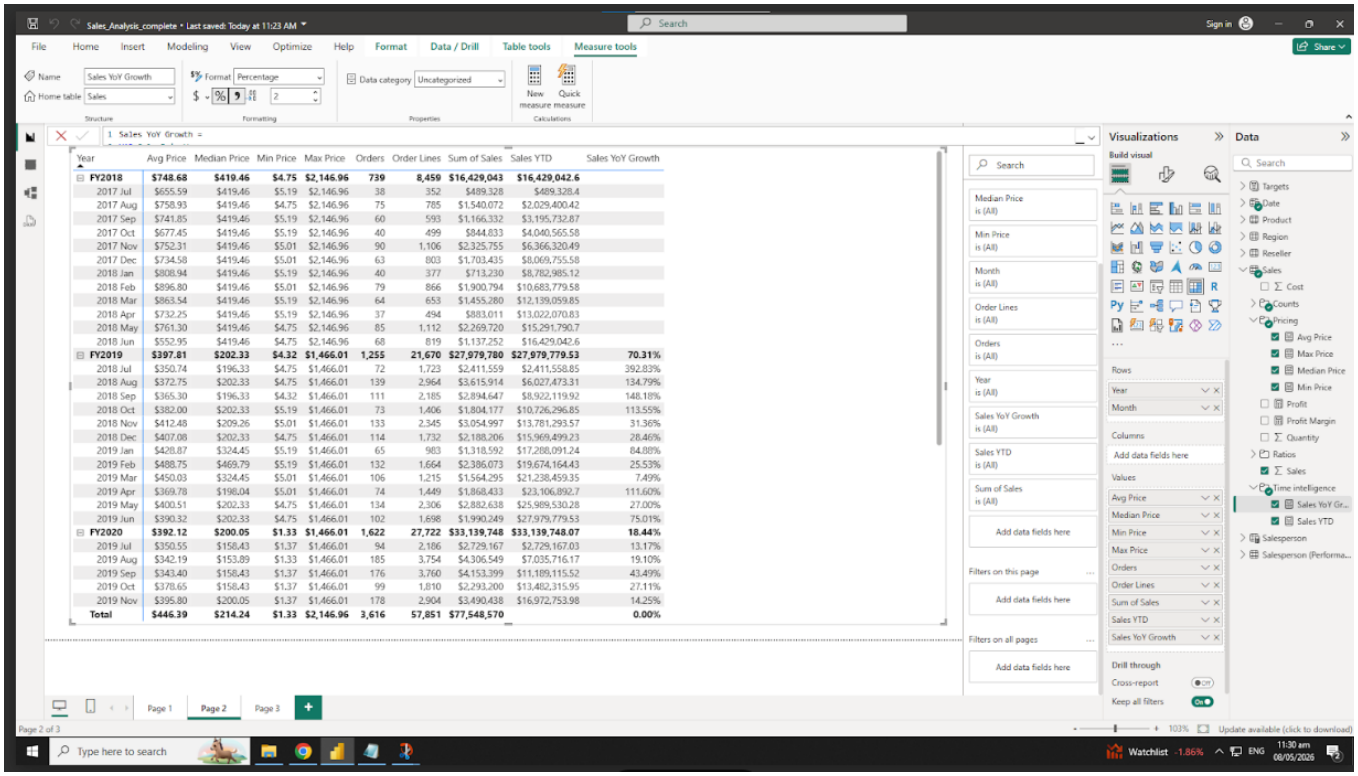
8. In **Model view**, place the two new measures into a display folder named *Time intelligence*.

- Time intelligence
  - Sales YoY Growth
  - Sales YTD

9. Save the Power BI Desktop file.

10. Save File as Sales_Analysis_complete.pbix 11. **Upload the pbix file and screenshot of your output and submit in gclass and post in your portfolio (with Task description and screenshot)**

Output:



Sales_Analysis_complete • Last saved: Today at 11:23 AM
File Home Insert Modeling View Optimize Help Format Data / Drill Table tools Measure tools
Name Sales YoY Growth
Home table Sales
Format Percentage
Data category Uncategorized
New measure Quick measure
1 Sales YoY Growth =
Year Avg Price Median Price Min Price Max Price Orders Order Lines Sum of Sales Sales YTD Sales YoY Growth
FY2018 $748.68 $419.46 $4.75 $2,146.96 739 8,459 $16,429,043 $16,429,042.6
2017 Jul $655.59 $419.46 $5.19 $2,146.96 38 352 $489,328 $489,328.4
2017 Aug $758.93 $419.46 $4.75 $2,146.96 75 785 $1,540,072 $2,029,400.42
2017 Sep $741.85 $419.46 $5.19 $2,146.96 60 593 $1,166,332 $3,195,732.87
2017 Oct $677.45 $419.46 $5.19 $2,146.96 40 499 $844,833 $4,040,565.58
2017 Nov $752.31 $419.46 $5.01 $2,146.96 90 1,106 $2,325,755 $6,366,320.49
2017 Dec $734.58 $419.46 $5.01 $2,146.96 63 803 $1,703,435 $8,069,755.58
2018 Jan $808.94 $419.46 $5.19 $2,146.96 40 377 $713,230 $8,782,985.12
2018 Feb $896.80 $419.46 $5.01 $2,146.96 79 866 $1,900,794 $10,683,779.58
2018 Mar $863.54 $419.46 $5.19 $2,146.96 64 653 $1,455,280 $12,139,059.85
2018 Apr $732.25 $419.46 $5.19 $2,146.96 37 494 $883,011 $13,022,070.83
2018 May $761.30 $419.46 $4.75 $2,146.96 85 1,112 $2,269,720 $15,291,790.7
2018 Jun $552.95 $419.46 $4.75 $2,146.96 68 819 $1,137,252 $16,429,042.6
FY2019 $397.81 $202.33 $4.32 $1,466.01 1,255 21,670 $27,979,780 $27,979,779.53 70.31%
2018 Jul $350.74 $196.33 $4.75 $1,466.01 72 1,723 $2,411,559 $2,411,558.85 392.83%
2018 Aug $372.75 $202.33 $4.75 $1,466.01 139 2,964 $3,615,914 $6,027,473.31 134.79%
2018 Sep $365.30 $196.33 $4.32 $1,466.01 111 2,185 $2,894,647 $8,922,119.92 148.18%
2018 Oct $382.00 $202.33 $5.19 $1,466.01 73 1,406 $1,804,177 $10,726,296.85 113.55%
2018 Nov $412.48 $209.26 $5.01 $1,466.01 133 2,345 $3,054,997 $13,781,293.57 31.36%
2018 Dec $407.08 $202.33 $4.75 $1,466.01 114 1,732 $2,188,206 $15,969,499.23 28.46%
2019 Jan $428.87 $324.45 $5.19 $1,466.01 65 983 $1,318,592 $17,288,091.24 84.88%
2019 Feb $488.75 $469.79 $5.19 $1,466.01 132 1,664 $2,386,073 $19,674,164.43 25.53%
2019 Mar $450.03 $324.45 $5.01 $1,466.01 106 1,215 $1,564,295 $21,238,459.35 7.49%
2019 Apr $369.78 $198.04 $5.01 $1,466.01 74 1,449 $1,868,433 $23,106,892.7 111.60%
2019 May $400.51 $202.33 $4.75 $1,466.01 134 2,306 $2,882,638 $25,989,530.28 27.00%
2019 Jun $390.32 $202.33 $4.75 $1,466.01 102 1,698 $1,990,249 $27,979,779.53 75.01%
FY2020 $392.12 $200.05 $1.33 $1,466.01 1,622 27,722 $33,139,748 $33,139,748.07 18.44%
2019 Jul $350.55 $158.43 $1.37 $1,466.01 94 2,186 $2,729,167 $2,729,167.03 13.17%
2019 Aug $342.19 $153.89 $1.33 $1,466.01 185 3,754 $4,306,549 $7,035,716.17 19.10%
2019 Sep $343.40 $158.43 $1.37 $1,466.01 176 3,760 $4,153,399 $11,189,115.52 43.49%
2019 Oct $378.65 $158.43 $1.37 $1,466.01 99 1,810 $2,293,200 $13,482,315.95 27.11%
2019 Nov $395.80 $200.05 $1.37 $1,466.01 178 2,904 $3,490,438 $16,972,753.98 14.25%
Total $446.39 $214.24 $1.33 $2,146.96 3,616 57,851 $77,548,570 0.00%
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Visualizations
Data